A Python-Based Instrument Control System for Adaptive Optics: Architecture and Applications in the CaNaPy–ALASCA Experiment

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Abstract

The Instrument Control System (ICS) is a Python-based software framework designed to coordinate and manage a wide range of opto-mechanical and electro-optical devices within advanced adaptive optics (AO) systems. Its architecture enables high-level control and automation of a complex hardware environment and serves as a central interface for system configuration and operation. ICS is a core element of the CaNaPy-ALASCA facility, a collaborative project led by ESO in cooperation with ESA and member-states institutions and companies. CaNaPy goals are pre-compensation of the laser guide star uplink, demonstrating wavefront sensing of the LGS with a PyWFS and exploring atmospheric tip-tilt retrieval directly from a laser guide star using the time-delay method ALASCA extends this concept to satellite optical feeder link by implementing uplink pre-compensation to ensure stable and accurate ground-to-space laser links under real atmospheric conditions. The ICS controls a number of critical components, including the Optical Ground Station (OGS) telescope, laser pointing cameras (LPCs), 589 nm and infrared lasers, beam expanders, motorised stages, and monitoring of environmental parameters as well as the status of wavefront sensors. It supports both manual and automated workflows through a dedicated GUI, enabling repeatable, parameter-driven experiment execution. The ICS also manages synchronisation with external systems such as the Real-Time Controller (RTC), with which it dynamically communicates to support adaptive loop closure and real-time experiment control, the Telescope Control System (TCS) and laser shutter control, ensuring coordinated and reliable operation during on-sky and laboratory testing. The system continuously logs configuration parameters and telemetry to facilitate performance evaluation and troubleshooting. This work presents the current implementation of the ICS, its role in recent integration campaigns, and its key contributions to the successful deployment of the CaNaPy- ALASCA AO systems.

Keywords: Python, based software framework, Graphical User Interface (GUI)

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